

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T30 · Functions

With Answers

Questions with full worked solutions

■ 50 pages

FAMILY MAP

Functions

F01	Interpret function notation and mapping	2 Q 2 marks
F02	Compose functions in the stated order	4 Q 9 marks
F03	Control domain and range	5 Q 11 marks
F04	Construct an inverse function	12 Q 48 marks

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$

$g(x) = 4x^2$

(a) Find $f(9)$

(1)

Worked Solution 09

Function substitution

UNIT 2

1 marks

Substitute

$$f(9) = 9 - \sqrt{9} = 9 - 3 = 6.$$

Final answer

$$f(9) = 6$$

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$
 $g(x) = 4x^2$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

Worked Solution 10

Composite function inequality

UNIT 2
3 marks**Compose the functions rigorously**

$$fg(x) = f(4x^2) = 9 - \sqrt{4x^2} = 9 - 2|x|.$$

Solve the inequality as written

$$9 - 2|x| < 0 \Rightarrow |x| > \frac{9}{2} \Rightarrow x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

Read the domain statement carefully

The printed $x \geq 0$ is the input domain of f . Since $g(x) = 4x^2 \geq 0$ for every real x , it does not remove the negative branch.

If the original variable is additionally restricted

If the original variable is also restricted to $x \geq 0$, then $|x| = x$, so the result reduces to $x > \frac{9}{2}$.

Final answer

$$\text{As written: } x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

$$\text{With an additional } x \geq 0 \text{ restriction: } x > \frac{9}{2}.$$

Question 07

4MA1-1H Q15

ORIGINAL QUESTION

4 marks

15 The function f is defined as

$$f: x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of x that cannot be included in any domain of the function f

.....
(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(3)

Worked Solution 07

Domain and inverse function

MODULAR UNIT 2

4 marks

Domain

$$x - 2 \neq 0 \Rightarrow x \neq 2.$$

Inverse

$$y = (3x + 1)/(x - 2) \Rightarrow x(y - 3) = 2y + 1 \Rightarrow f^{-1}(x) = \frac{2x + 1}{x - 3}.$$

Final answer

$$x \neq 2; \quad f^{-1}(x) = \frac{2x + 1}{x - 3}$$

Question 31

4MA1-2H Q19 (a)

ORIGINAL QUESTION UNIT 2

1 marks

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(a) State which value of x cannot be included in any domain of g

.....
(1)

Worked solution 31

Excluded value from a rational function domain

MODULAR UNIT 2

1 marks

(a) – Excluded value from a rational function domain

Set the denominator to zero

$$2x - 9 = 0 \Rightarrow x = \frac{9}{2} = 4.5.$$

Final answer $x = 4.5$ cannot be included

Question 32

4MA1-2H Q19 (b)

ORIGINAL QUESTION UNIT 2

2 marks

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(b) Find $fg(4)$

.....
(2)

Worked solution 32

Composite function value

MODULAR UNIT 2

2 marks

(b) – Composite function valueEvaluate $g(4)$

$$g(4) = \frac{5}{2(4) - 9} = -5.$$

Apply f

$$fg(4) = f(-5) = 5(-5) + 7 = -18.$$

Final answer

-18

Question 33

4MA1-2H Q19 (c)

ORIGINAL QUESTION UNIT 2

4 marks

The function h is such that

$$h:x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function h^{-1} in the form $h^{-1}:x \mapsto \dots$

$$h^{-1}:x \mapsto \dots\dots\dots$$

(4)

Worked solution 33

Find the inverse of a restricted quadratic

MODULAR UNIT 2

4 marks

(c) – Find the inverse of a restricted quadratic

Complete the square

$$y = 3x^2 - 12x + 8 = 3(x - 2)^2 - 4.$$

Rearrange

$$y + 4 = 3(x - 2)^2 \Rightarrow (x - 2)^2 = \frac{y + 4}{3}.$$

Use the restriction

$$x > 2 \Rightarrow x - 2 > 0, \text{ so take the positive square root.}$$

Swap variables

$$h^{-1}(x) = 2 + \sqrt{\frac{x + 4}{3}}.$$

Final answer

$$h^{-1}(x) = 2 + \sqrt{(x+4)/3}$$

Question 14

4MA1-1H Q17

ORIGINAL QUESTION

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....
(1)

(b) Solve $gh(x) = 1$

.....
(3)

Worked Solution 14

Domain exclusion and composite equation

MODULAR UNIT 2

4 marks

Exclude a zero denominator

$$2x - 5 \neq 0 \Rightarrow x \neq \frac{5}{2}.$$

Form the composite

$$g(h(x)) = \frac{11}{2(x^2 + 4) - 5} = 1.$$

Solve with the stated nonnegative domain

$$2x^2 + 3 = 11 \Rightarrow x^2 = 4, \quad x = 2.$$

Final answer

$$x \neq \frac{5}{2}; \quad x = 2 \text{ for } g(h(x)) = 1$$

Original question 17

4MA1-1H Q25 M01

MODULAR UNIT 2

4 marks

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$$f^{-1}(x) = \dots\dots\dots$$

Worked solution 17

Find the inverse function

MODULAR UNIT 2

4 marks

M01 – Find the inverse function**Complete the square**

$$f(x) = 2x^2 - 24x + 7 = 2(x - 6)^2 - 65.$$

Rearrange

$$y + 65 = 2(x - 6)^2 \Rightarrow (x - 6)^2 = (y + 65)/2.$$

Use the domain $x \geq 6$

$$x = 6 + \sqrt{(y + 65)/2}.$$

Swap x and y

$$f^{-1}(x) = 6 + \sqrt{(x + 65)/2}.$$

Final answer

$$f^{-1}(x) = 6 + \sqrt{\frac{x + 65}{2}}$$

Original question 13

4MA1-1H Q14 Q14(a)

ORIGINAL QUESTION**1 marks****14** The function f is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(a) Find $f(10)$

(1)

Worked solution 13

Chapter: Functions

MODULAR UNIT 2

1 marks

Q14(a) – Chapter: Functions

Family: Interpret function notation and mapping • Variant: Evaluate a function at a numerical input

Method

$$f(10) = 2(10)/(10 - 6) = 20/4 = 5.$$

*Why: Substitute $x = 10$ into the function definition.***Final answer****f(10): 5**

Original question 14

4MA1-1H Q14 Q14(b)

ORIGINAL QUESTION**3 marks**

14 The function f is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(b) Express the inverse function f^{-1} in the form $f^{-1}: x \mapsto \dots$

$$f^{-1}: x \mapsto \dots \dots \dots (3)$$

Worked solution 14

Chapter: Functions

MODULAR UNIT 2

3 marks

Q14(b) – Chapter: Functions

Family: Construct an inverse function • Variant: Inverse of a linear or fractional function

MethodLet $y = 2x/(x - 6)$; then $y(x - 6) = 2x$.*Why: Introduce y and clear the denominator.***Method** $yx - 6y = 2x$, so $x(y - 2) = 6y$ and $x = 6y/(y - 2)$.*Why: Collect the terms containing x and factor them.***Method**Interchanging the input and output letters gives $f^{-1}(x) = 6x/(x - 2)$.*Why: The inverse reverses the mapping between x and y .***Final answer****inverse function:** $f^{-1}(x) = 6x/(x - 2)$

Original question 36

4MA1-2H Q25 Q25(a)

ORIGINAL QUESTION**4 marks**

25 The function g is defined as

$$g:x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x:x \geqslant 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1}:x \mapsto \dots$

$$g^{-1}:x \mapsto \dots \quad (4)$$

Worked solution 36

Chapter: Functions

MODULAR UNIT 2

4 marks

Q25(a) – Chapter: Functions

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Method

$$g(x) = 5 + 6x - x^2 = 14 - (x-3)^2.$$

*Why: Complete the square.***Method**

$$\text{For } x \geq 3, y = 14 - (x-3)^2, \text{ so } (x-3)^2 = 14 - y.$$

*Why: Use the restricted domain to choose the positive branch.***Method**

$$x-3 = \sqrt{(14-y)}.$$

*Why: Take the positive square root.***Method**

$$\text{Therefore } g^{-1}(x) = 3 + \sqrt{(14-x)}.$$

*Why: Swap x and y to write the inverse function.***Final answer**

$$\text{inverse: } g^{-1}(x) = 3 + \sqrt{(14-x)}$$

Original question 37

4MA1-2H Q25 Q25(b)

ORIGINAL QUESTION**1 marks**

(b) State the domain of g^{-1}

.....
(1)

Worked solution 37

Chapter: Functions

MODULAR UNIT 2

1 marks

Q25(b) – Chapter: Functions

Family: Control domain and range • Variant: Domain and range swap under inversion

Method

The square root requires $14-x \geq 0$, so the domain of g^{-1} is $x \leq 14$.

Why: Use the radicand condition.

Final answer**domain:** $x \leq 14$

21 The functions f and g are such that

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

The function h is such that $h(x) = fg(x)$ for $x \geq -2$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

Worked solution

January 2020 | Question 21 | 5 marks

Family: Construct an inverse function | Variant: Inverse of a composite-defined function

METHOD / WORKING

$$1. g(x) = x + 3, \text{ so } f(g(x)) = (x + 3)^2 - 2(x + 3) = x^2 + 4x + 3.$$

$$2. \text{ Complete square: } h(x) = (x+2)^2 - 1, \text{ with } x \geq -2.$$

$$3. \text{ Let } y = (x+2)^2 - 1.$$

$$4. \text{ Then } (x+2)^2 = y+1 \text{ and } x+2 = \sqrt{y+1} \text{ on the restricted domain.}$$

$$5. \text{ Swap } x, y: h^{-1}(x) = \sqrt{x+1} - 2.$$

FINAL ANSWER

$$h^{-1}(x) = \sqrt{x+1} - 2$$

22 The function f is such that $f(x) = x^2 - 8x + 5$ where $x \leq 4$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

Worked solution

January 2021 | Question 22 | 3 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Method / working

1. Complete the square: $f(x) = (x-4)^2 - 11$ with $x \leq 4$.
 2. Set $y = (x-4)^2 - 11$ and choose $x-4 = -\sqrt{y+11}$ because $x \leq 4$.
 3. Swap x and y : $f^{-1}(x) = 4 - \sqrt{x+11}$.
- Final answer: $f^{-1}(x) = 4 - \sqrt{x+11}$

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

Worked solution

June 2021 | Question 24 | 2 marks

Family: Compose functions in the stated order • Variant: Evaluate a numerical composite function

Method / working

1. $g(2) = 7 \times 2 - 6 = 8.$

2. $f(g(2)) = f(8) = 5(8)^2 - 10(8) + 7 = 247.$

Final answer: 247

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

Worked solution

June 2021 | Question 24 | 4 marks

Family: Construct an inverse function | Variant: Inverse of a restricted quadratic function

METHOD / WORKING

1. Write $y = 5x^2 - 10x + 7 = 5(x - 1)^2 + 2$ with $x \geq 1$.
2. Rearrange: $(x-1)^2 = (y-2)/5$.
3. Use the domain $x \geq 1$, so $x-1$ is the positive square root.
4. Swap x and y : $f^{-1}(x) = 1 + \sqrt{(x-2)/5}$.

FINAL ANSWER

$$f^{-1}(x) = 1 + \sqrt{(x-2)/5}$$

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$g^{-1}(x) =$
(3)

Worked solution

November 2021 | Question 15 | 1 marks

Family: Control domain and range | Variant: Exclude a denominator zero from the domain

METHOD / WORKING

$$1. f(x) = 2x - 3, g(x) = \frac{x}{3x + 1}$$

FINAL ANSWER

Excluded value $-\frac{1}{3}$, gf

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$g^{-1}(x) =$
(3)

Worked solution

November 2021 | Question 15 | 2 marks

Family: Compose functions in the stated order | Variant: Derive a composite function rule

METHOD / WORKING

1. (a) The denominator of g cannot be zero:

2. $3x+1=0$

3. $x=-\frac{1}{3}$

FINAL ANSWER

$$(x) = \frac{2x-3}{6x-8}, \text{ and } g^{-1}$$

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$g^{-1}(x) =$
(3)

Worked solution

November 2021 | Question 15 | 3 marks

Family: Construct an inverse function | Variant: Inverse of a linear or fractional function

METHOD / WORKING

1. (b)

2. $gf(x)=g(f(x))$

3. $gf(x) = \frac{2x-3}{3(2x-3)+1}$

4. $gf(x) = \frac{2x-3}{6x-8}$

FINAL ANSWER

$$(x) = \frac{x}{1-3x}$$

21 The function f is such that $f(x) = 5 + 6x - x^2$ for $x \leq 3$

(a) Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

.....
(2)

(b) Using your answer to part (a), find the range of values of x for which $f^{-1}(x)$ is positive.

.....
(5)

.....
(Total for Question 21 is 7 marks)

Answer reference | May-Nov 2020 | Question 21

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

21	a	$5 - (x \pm q)^2 + 9$ oe or $p - (x - 3)^2$ oe or $p - q^2 + 2qx - x^2$ and one of $2q = 6$ or $p - q^2 = 5$			M1 may be seen in working eg $-(x - 3)^2 - 9 - 5]$ or expanding $p - (x - q)^2$ correctly and equating one of the coefficient of x or the constant term
			$14 - (x - 3)^2$	2	A1 fully correct SCB1 for $(x - 3)^2 - 14$
	b	e.g. $(x - 3)^2 = 14 - y$ [or $(y - 3)^2 = 14 - x]$			M1 correct steps to isolate their bracket fit from (a) dep on expression in form $\pm p \pm (x - q)^2$
		$x = 3 \pm \sqrt{14 - y}$ [or $y = 3 \pm \sqrt{14 - x}$]			M1 complete method to find y in terms of x or x in terms of y . Condone + for $\pm$ fit from (a) dep on expression in form $\pm p \pm (x - q)^2$
		$(f^{-1}(x) =) 3 - \sqrt{14 - x}$			M1 for the correct inverse
					M1 method to solve $0 < 3 - \sqrt{14 - x}$ or a lower bound of 5 clearly shown, eg $x > 5$ as part of the answer
			$5 < x \leq 14$	5	A1 cao
Total 7 marks					

Worked solution

May-Nov 2020 | Question 21 | 7 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Official final mark-scheme pages are embedded immediately after this question.

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

Answer reference | January 2023 | Question 18

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

18	(a)		$\frac{k}{x}$	1	B1 allow kx^{-1}
	(b)(i)		-46	1	B1 cao
	(ii)	$\frac{3(2-3x^4)}{2-(2-3x^4)}$ or $\frac{6-9x^4}{2-2+3x^4}$ oe or $\frac{6-9x^4}{3x^4}$ oe		2	M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{2-3x^4}{x^4}$		A1 allow $\frac{2}{x^4} - 3$ oe
					Total 4 marks

Worked solution

January 2023 | Question 18 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

Official final mark-scheme pages are embedded immediately after this question.

25 The function f is such that $f(x) = 3x^2 - 12x + 7$ where $x \leq 2$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

(Total for Question 25 is 4 marks)

Answer reference | January 2023 | Question 25

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Q	working	Answer	Mark	Notes
25	$y = 3(x^2 - 4x) + 7$ or $y = 3\left(x^2 - 4x + \frac{7}{3}\right)$ or $\frac{y-7}{3} = x^2 - 4x$ or $y = 3(x-2)^2 \dots$		4	M1 for a correct equation for a first step to complete the square
	eg $y = 3((x-2)^2 - 2^2) + 7$ or $y = 3((x-2)^2 - 2^2 + \frac{7}{3})$ or $y = 3(x-2)^2 - 5$ oe or			M1
	$(x-2)^2 = \frac{y+5}{3}$ oe eg $(x-2)^2 = \frac{y-7}{3} + 4$ or $x-2 = (\pm)\sqrt{\frac{y+5}{3}}$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	$2 - \sqrt{\frac{x+5}{3}}$		A1 oe NB: note only negative square root. Must be in terms of x any equivalent form
	Note: Allow candidates to swap x and y when finding the inverse			Total 4 marks
25 Alt	$3x^2 - 12x + (7 - y) = 0$		4	M1 for a correct first step
	$(x =) \frac{12 \pm \sqrt{144 - 12(7 - y)}}{6}$			M1
	$(x =) 2 \pm \sqrt{\frac{60 + 12y}{36}}$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	$2 - \sqrt{\frac{x+5}{3}}$		A1 oe NB: note only negative square root. Must be in terms of x any equivalent form
				Total 4 marks

Worked solution

January 2023 | Question 25 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Official final mark-scheme pages are embedded immediately after this question.

19 The functions f and g are such that

$$f(x) = 3x - 2$$

$$g(x) = \frac{x}{2x - 1}$$

(a) Find $g(3)$

.....
(1)

(b) Find $gf(x)$

Give your answer in its simplest form.

$gf(x) =$
(2)

(Total for Question 19 is 3 marks)

Worked solution

November 2025 | Question 19 | 3 marks

Family: Construct an inverse function | Variant: Inverse of a linear or fractional function

METHOD / WORKING

Worked solution
November 2025 Question 19 3 marks
Family: Construct an inverse function Variant: Inverse of a linear or fractional function
Solution
$f(x)=3x-2$, and $g(x)=(x)/(x-1)$
(a)
$g(3)=(3)/(3-1)=\frac{3}{2}$
(b)
$gf(x)=g(f(x))$
$gf(x)=(3x-2)/((3x-2)-1)$
$gf(x)=(3x-2)/(3x-3)$
Answer : $g(3) = \frac{3}{2}$, and $gf(x) = \frac{3x-2}{3x-3}$.
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER